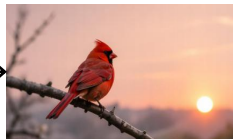


Prompt
&
Candidate
Generations

Prompt:
A red bird on a
branch at sunrise



Candidate A

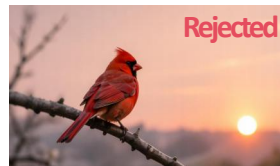


Candidate B

Human
Preference
Pair



Human
Pairwise
Comparison



Diffusion-DPO
Objective

DPO objective for diffusion
ELBO-based likelihood
reformulation

$$\log p_{\theta}(x_{\text{preferred}}) > \log p_{\theta}(x_{\text{rejected}})$$

Optimize Diffusion Model



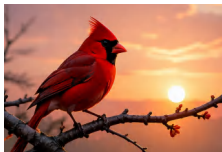
x_t



Model



x_0



Alignment toward preferred